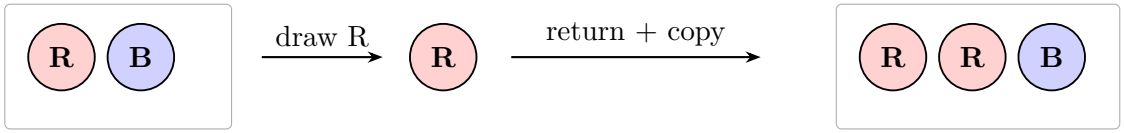
Start with one red and one blue. Draw, return it, add one counter of that color, and mix. For random play, each counter has the same chance; for possible stories, choose the draws yourself.



Marks distinguish counters of the same color. Start with R1 and B1. The first added counter gets mark 2; the next gets mark 3.



draw R1 return it and add R2 next draw could be R2

Problem 1: Find all the two-draw marked histories. Use them to decide whether zero, one, or two red draws has the greatest chance, if any.

Problem 2: Keep the starting bag R1, B1, but return each draw without adding a copy. How does the chance of zero, one, or two red draws change? Explain the difference exactly.

Problem 3: Return to copying. Compare the color stories RRB, RBR, and BRR. Does their order change their chance? Give an exact reason for your answer.

RRB

RBR

BRR